

KHABAROVSK, Russian Federation

WMO: 317350

Lat: **48.528N**Lon: **135.188E**Elev: **76**StdP: **100.42**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-28.1	-33.9	0.2	-29.1	-32.0	0.2	-27.5	12.3	-13.7	10.9	-18.3	1.8	240	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	30.2	22.6	28.8	21.8	27.1	21.0	24.2	27.8	23.2	26.7	22.3	25.7	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.0	25.7	22.1	16.9	24.9	21.0	15.8	24.2	73.6	27.5	69.5	26.4	66.0	25.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.0	9.6	8.5	-33.2	32.9	2.9	2.0	-35.3	34.4	-36.9	35.6	-38.5	36.7	-40.6	38.2	28.6
(5)				-33.3	25.8	2.8	1.1	-35.4	26.6	-37.0	27.2	-38.6	27.8	-40.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-20.4	-15.8	-6.7	4.6	12.7	18.1	21.7	20.1	14.0	5.1	-7.4	-18.2
	DBStd	15.49	4.77	4.81	5.48	4.59	4.13	3.60	2.83	2.80	3.66	4.38	6.23	4.87
	HDD10.0	3943	943	724	517	174	21	0	0	0	8	161	523	873
	HDD18.3	6044	1201	957	775	413	181	47	6	14	134	410	773	1132
	CDD10.0	1169	0	0	0	11	103	244	361	312	129	9	0	0
	CDD18.3	228	0	0	0	0	5	41	109	68	6	0	0	0
	CDH23.3	1741	0	0	0	5	79	380	821	414	41	1	0	0
	CDH26.7	428	0	0	0	1	12	101	241	71	3	0	0	0
Wind	WSAvg	3.7	3.3	3.7	4.2	4.1	3.9	3.3	2.9	3.0	3.4	4.2	4.7	4.0
Precipitation	PrecAvg	673	13	11	19	41	61	76	135	140	89	49	23	16
	PrecMax	1099	35	32	45	130	146	166	301	435	234	126	70	42
	PrecMin	378	0	0	1	0	16	2	7	25	14	1	3	2
	PrecStd	137	9	8	11	23	28	41	76	78	51	34	14	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.9	1.0	9.9	23.1	28.2	31.9	33.2	30.4	26.6	21.1	9.1	-1.8
		MCWB	-7.0	-1.9	3.9	12.4	17.3	21.9	23.2	24.4	19.7	13.5	4.8	-3.2
	2%	DB	-8.9	-2.6	6.2	19.1	25.3	29.2	31.1	28.5	24.2	17.1	6.0	-6.1
		MCWB	-9.6	-4.7	1.7	9.9	15.1	20.7	23.0	23.0	18.3	11.1	3.2	-7.1
	5%	DB	-11.1	-5.8	3.7	15.2	23.2	27.2	29.2	27.1	22.8	14.2	3.8	-9.0
		MCWB	-11.8	-7.3	0.4	7.9	14.0	19.7	22.4	21.9	17.3	9.2	1.3	-9.7
	10%	DB	-13.1	-8.2	1.5	12.6	21.0	25.3	27.7	25.9	20.8	12.0	1.2	-11.0
		MCWB	-13.8	-9.3	-1.0	6.7	13.3	19.0	22.0	21.0	16.2	7.9	-0.5	-11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.9	-1.8	4.2	12.9	18.5	23.9	25.5	26.0	21.1	14.3	6.1	-3.0
		MCDB	-6.1	0.5	8.5	21.8	25.5	29.2	29.6	28.9	24.7	18.9	8.6	-1.8
	2%	WB	-9.6	-4.8	2.4	10.6	16.6	21.9	24.2	24.4	19.7	12.0	3.3	-7.1
		MCDB	-8.9	-2.8	5.5	17.3	23.1	27.0	28.4	26.7	22.8	16.2	5.5	-6.2
	5%	WB	-11.9	-7.1	0.6	8.8	15.3	20.7	23.4	23.1	18.3	10.2	1.3	-9.9
		MCDB	-11.3	-5.7	3.1	14.7	21.0	25.9	27.5	25.7	21.4	13.1	3.2	-9.2
	10%	WB	-13.9	-9.4	-0.9	7.1	14.1	19.7	22.6	22.1	16.9	8.4	-0.3	-11.8
		MCDB	-13.3	-8.4	1.4	11.7	19.6	24.2	26.4	24.8	19.8	11.5	1.2	-11.2
Mean Daily Temperature Range		MDBR	8.4	9.6	9.5	10.4	11.4	10.6	9.5	8.8	10.2	9.6	7.6	7.4
	5% DB	MCDBR	9.1	11.0	10.6	14.7	15.4	13.3	12.1	10.9	12.5	13.0	9.6	8.4
		MCWBR	8.4	9.5	7.3	8.1	7.4	6.2	5.5	5.8	6.9	8.0	7.4	7.7
		MCDBR	9.1	10.9	10.0	13.8	13.1	11.7	10.3	9.2	10.6	11.8	8.8	8.1
	5% WB	MCWBR	8.4	9.4	7.3	8.2	7.2	6.1	5.4	5.6	6.6	8.1	7.1	7.6
Clear-Sky Solar Irradiance	taub	0.287	0.332	0.406	0.532	0.559	0.515	0.503	0.448	0.413	0.438	0.370	0.301	
	taud	2.089	1.954	1.833	1.746	1.776	1.986	2.063	2.224	2.224	1.979	2.020	2.062	
	Ebn at Noon	747	784	778	718	720	761	763	792	779	654	618	661	
	Edh at Noon	87	128	172	209	211	172	158	129	116	126	96	80	
All-Sky Solar Radiation	RadAvg	1.87	3.12	4.57	4.89	5.35	5.69	5.27	4.66	3.83	2.60	1.69	1.44	
	RadStd	0.08	0.10	0.28	0.47	0.45	0.69	0.46	0.30	0.29	0.18	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page